Mathematics - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. 5.2- II. Evaluate $\int_0^{0.4} \frac{\Gamma(7)}{\Gamma(4)\Gamma(3)} y^3(1-y)^2 dy\$$ (a) Using integration.
   (a) 0.3087
   (b) 0.2148
   (c) 0.0896
   (d) 0.3456
   (e) 0.0991
2. Two numbers added together are 19. Their product is 70. What are the two numbers?
   (a) 5, 14
   (b) 4, 15
   (c) 1, 18
   (d) 9, 10
   (e) 12, 7
3. Passing to polar coordinates, calculate the double integral $\iint_S ydxdy$ with $y > 0$, where S is a semicircle of a diameter 1 with center at point C(1/2, 0) above the X axis.
   (a) 0.333
   (b) 0.1666
   (c) 0.5
   (d) 0.125
   (e) 0.625
4. A number, rounded to the nearest thousand,is 47,000. Which number could be the number that was rounded?
   (a) 46,504
   (b) 46,295
   (c) 47,924
   (d) 47,999
   (e) 47,520
5. Write 7 over 33 as a decimal.
   (a) 0.021 repeating

(b) 0.3333

(c) 0.23 Repeating

(d) 0.21 Repeating

(e) 0.07 Repeating

True / False (10 marks)

Answer True or False

6. True or False: If Owen's waist circumference increases by 15%, his new jean waist size will be 36 inches.
7. True or False: The expression $1990 \times 1991 - 1989 \times 1990$ simplifies to the counting number 3980.
8. True or False: The mixed number representation of 1 and 1 over 11 is equivalent to 1.1 over 11 in simplest form.
9. True or False: There are exactly 36 distinct triangles with integer side lengths that have a maximum side length of 11.
10. True or False: The sum of -9 and -8 is -17, as both numbers are negative and their absolute values add together.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the sum of the roots, real and non-real, of the equation $x^{2001} + \left(\frac{1}{2} - x\right)^{2001} = 0$, given that there are no multiple roots.
12. Using the binomial theorem, derive the coefficient of x^2y^5 in the expansion of $(x+2y)^7$ and discuss the significance of your result in combinatorial contexts.

End of Paper